Local Proof Architectures for Critical PDE Regularity

State-Space Stratification, Fault Containment, and LLM-Auditable Proof Design

Guillem Duran Ballester

`[email / affiliation placeholder]`

Working draft — April 26, 2026

Abstract

This paper proposes a verification-aware proof architecture for critical PDE regularity problems based on local state-space stratification. The framework replaces a monolithic contradiction proof by a finite directed graph of proof capsules, each with explicit hypotheses, outputs, retained obstructions, failure routes, and audit tests. A candidate singular branch is routed through scale-invariant local tests until it reaches either a closed terminal stratum or a named open obstruction. By isolating the singularity into compact local strata, the architecture bypasses the need for global critical-norm estimates or whole-space Liouville theorems: nonlocal pressure tails at spatial infinity are walled off, and local Calderón–Zygmund control on compact cylinders becomes sufficient for the active branch. The framework treats a PDE proof as a fault-tolerant software system whose logical structure is an abstract syntax tree suitable for direct LLM processing and eventual formal verification in systems such as Lean 4. The three-dimensional incompressible Navier–Stokes equations are used as a case study, with Caffarelli–Kohn–Nirenberg concentration, Type I/Type II separation, ancient-profile routing, repaired gauges, scale-cost mechanisms, and terminal profile exits serving as motivating examples.

Status of this draft. This document is a paper skeleton: every section has the final intended role, but most entries are written as one-sentence bullets or theorem templates to be expanded later.

Core principle. The paper studies proof architecture and verification structure by isolating every analytic step into an independently auditable capsule recorded in an explicit ledger.

Contents

1 Introduction

1.1 Motivation: why monolithic PDE proofs are fragile

- Critical PDE regularity arguments often fail for architectural reasons: a single misunderstood compactness limit or pressure normalization can silently contaminate dozens of later pages.
- Monolithic contradiction proofs force local facts, global estimates, and rigidity arguments into one fragile trunk, so one bad step on page 84 can snap the entire proof.
- The real bottleneck is not only human memory; it is a proof format that allows hidden global dependencies to leak across the manuscript without typed interfaces.
- LLM-assisted proof search exposes this architectural flaw even more sharply, because a model can verify a local derivation while still missing the global estimate leak that makes the overall argument unsound.

1.2 The Curse of Globality

- Classical regularity programs often try to impose global constraints such as whole-space critical-norm bounds on equations whose dynamics still carry nonlocal terms, most notably pressure.

- Those global bounds accumulate interacting cutoff errors, pressure tails, and compactness losses across many scales until the bookkeeping itself becomes the dominant difficulty.
- A proof built around global L^3 control, global profile decompositions, or whole-space Liouville inputs is therefore structurally predisposed to catastrophic coupling between distant parts of the argument.
- The curse of globality is that every new estimate must remain compatible with the entire ambient space at once, even when the singularity mechanism is fundamentally local.

1.3 The Modular/Local Paradigm

- The proposed alternative is to replace global proof trunks by finite state-space dichotomies driven only by local, scale-invariant observables on compact parabolic cylinders.
- In this paradigm, global estimates are a relic of monolithic proof structure rather than a mathematical necessity: each branch only needs the local bounds generated inside its own state.
- The resulting proof is fault-tolerant and strictly bounded, because every local estimate has a finite area of effect and every branch can be audited without importing the entire ambient space.
- This is the same conceptual move that makes modern software reliable: isolate modules, define interfaces, and forbid undeclared dependencies from leaking across the system.

1.4 The proposed alternative

- We organize a regularity proof as a finite routed state space rather than as one uninterrupted global argument.
- Each edge in the state-space graph is a proof capsule with a strict local contract: inputs, outputs, retained invariant, failure routes, and audit obligations.
- The architecture compiles a continuous infinite-dimensional PDE problem into a discrete mathematical API whose nodes and edges can be checked one local interface at a time.
- A failure of a capsule does not erase the whole architecture; it reopens one edge or terminal node and leaves the verified subgraph meaningful.

1.5 Main contributions

- We define proof capsules, retained obstructions, routing graphs, terminal closures, and fault containment.
- We elevate elimination of hidden global estimates to a design principle: every branch is expressed through local observables, local gauges, and local closure capsules.
- We prove an abstract routed-closure theorem: if every branch is routed and every terminal node is closed against a retained obstruction, then the root branch is impossible.
- We propose an LLM-audit protocol for scaling, sign, pressure, compactness, dependency, and failure-route checks, tuned to the local reasoning window of current models.

- We introduce a mathematical API that compiles continuous, infinite-dimensional PDE dynamics into discrete local nodes that LLMs and formal systems can navigate without exponential blow-up of the search space.
- We give the Navier–Stokes singularity-routing program as a detailed case study and stress test for machine-auditable proof design, with the routing graph functioning as the proof's logical AST.

Takeaway. The novelty is not a single new estimate; the novelty is a proof architecture in which every estimate is local, every failure of an estimate has a semantic destination, and every terminal contradiction is tied to a retained obstruction.

1.6 Relationship with current literature

- A template for concentration compactness arguments, but with the concentration mechanism promoted from one lemma inside a long proof to the front door of an explicit routing graph.
- Similarities to profile decomposition and Littlewood–Paley methods, except that those tools are deployed only on specific strata rather than across the entire state space at once.
- Compatibility with partial regularity theory and imported rigidity theorems, which now appear as typed routing endpoints rather than as diffuse background assumptions.

2 Overcoming the Global Estimate Bottleneck

2.1 Analytic surgery instead of global smoothness enforcement

- The guiding analogy is Perelman's Ricci flow surgery: instead of demanding that the full evolution remain globally well behaved in one stroke, we isolate the dangerous region and classify the singular behavior into finitely many irreducible atoms.
- In the Navier–Stokes setting those atoms are compact cores, rough cores, multibubbles, cascades, terminal profiles, and related local descendants.
- The proof architecture is therefore surgical rather than monolithic: singular behavior is not denied abstractly, but routed into a finite taxonomy where each atom has its own closure capsule.

2.2 Walling off spatial infinity

- The architecture works on compact parabolic cylinders with explicit cutoff functions, so only the immediate area of effect of the candidate singularity enters the active calculation.
- Once the proof is routed into a local stratum, spatial infinity is walled off rather than globally integrated into every later estimate.
- This is the structural reason the program can discard whole-space critical-norm control: the branch only pays for the geometry that survives inside its local window.

2.3 Covariant gauge observers and local pressure control

- The pressure term is handled by local gauge fixing and covariant observers adapted to the active branch rather than by a single global reconstruction formula imposed on all scales at once.
- Repaired gauges flatten the local connection enough to expose explicit modulation coefficients, while local Calderón–Zygmund estimates control the pressure contribution inside the compact cylinder actually used by the branch.
- This local observer viewpoint prevents pressure tails at infinity from leaking back into every estimate as a hidden global error source.

2.4 Why this geometry matches machine reasoning

- LLMs and formal tactic engines struggle when asked to reason over a large topological proof trunk whose active hypotheses are spread across many global layers.
- They perform far better on compact local tasks such as a cutoff estimate, a local Sobolev embedding, a Calderón–Zygmund bound, or a compactness passage inside a fixed cylinder.
- The architecture is therefore mathematically stronger and computationally better aligned: it redesigns the proof so that the dominant reasoning objects already fit the cognitive profile of machine auditors.

3 Abstract Routed Proof Architectures

3.1 Branches, states, and observables

Definition 3.1 (Branch)**.** A branch is a candidate counterexample trajectory together with the local data retained after all preceding routing steps.

Definition 3.2 (State)**.** A state is a named local regime characterized by scale-invariant observables, compactness data, and declared failure alternatives.

- Examples of observables include critical concentration, local energy, pressure oscillation, terminal mass, scale drift, profile count, and tail defect.
- States are defined purely by local, scale-invariant observables on compact windows; they do not depend on boundary data at spatial infinity.
- A state should be stable under the rescalings, gauges, and subsequences used before its closure theorem is invoked.

3.2 Retained obstructions

Definition 3.3 (Retained obstruction)**.** A retained obstruction is a nonzero scale-invariant quantity attached to a branch and required to survive every allowed extraction, rescaling, gauge change, or descendant operation.

- The retained obstruction is the object contradicted by terminal rigidity or vanishing.
- Typical examples are positive CKN concentration, compact local L^3 mass, nonzero terminal profile mass, or a nonvanishing localized core.

3.3 Proof capsules

Definition 3.4 (Proof capsule)**.** A proof capsule is a tuple

$$\mathfrak{C} = (H, O, I, F, A),$$

where H is the input contract, O is the output contract, I is the retained obstruction, F is the list of named failure routes, and A is the audit checklist.

- A proof capsule is a mathematical API contract: H is the pre-condition, O is the post-condition, and the retained obstruction plus failure routes specify what is preserved if the edge succeeds or how the branch is retyped if it does not.
- This strict typing is what prevents hidden global-estimate leaks: every theorem invocation must expose the exact local data it consumes and the exact local data it exports.

Capsule field	Meaning
Input contract H	Exact hypotheses the capsule is allowed to use.
Output contract O	The local conclusion produced if the capsule succeeds.
Retained obstruction I	The invariant that must survive the capsule.
Failure routes F	Named alternatives entered if the capsule hypotheses fail.
Audit checklist A	Scaling, sign, compactness, pressure, dependency, and locality tests.

3.4 Routing graphs

Definition 3.5 (Routing graph)**.** A routing graph is a finite directed graph $G = (V, E)$ with a root state, edge labels given by proof capsules, and terminal nodes marked either closed or open.

- Edges are local theorems or routing lemmas.
- Terminal nodes are either contradictions, lower already-closed strata, or explicitly open obstructions.
- A graph is useful even before every terminal node is closed, because it records exactly what remains to be proved.

3.5 The abstract routed-closure principle

Theorem 3.6 (Abstract routed-closure principle)**.** *Let G be a finite rooted routing graph. Suppose every root branch enters at least one outgoing edge, every edge is justified by a valid proof capsule whose hypotheses are produced upstream, every retained obstruction is preserved along edges, and every terminal node is closed by contradiction with its retained obstruction. Then no root branch exists.*

Proof sketch. Follow any hypothetical root branch through the finite graph; exhaustiveness gives a terminal node, preservation gives a surviving obstruction there, and terminal closure contradicts that obstruction. □

3.6 Fault containment

Definition 3.7 (Fault containment)**.** A routed proof architecture has fault containment if invalidating one capsule changes only the closure status of the nodes and paths depending on that capsule, while leaving the rest of the graph meaningful.

- Every capsule has an area of effect: the topological region of the routing graph whose correctness genuinely depends on that local estimate.
- Fault containment is the demand that an error remain quarantined inside its area of effect rather than poisoning the proof trunk globally.

Proposition 3.8 (Local invalidation principle)**.** *If a capsule $e \in E(G)$ is invalidated, then all branches not passing through e remain governed by the verified subgraph $G \setminus \{e\}$.*

Proof sketch. The proof is graph-theoretic: dependency is path-local, so invalidating an edge affects exactly those terminal claims whose proof paths use that edge. □

4 Auditability and LLM-Oriented Proof Design

4.1 Common failure modes of LLM-generated proofs

- Monolithic proofs exceed the active reasoning window of current language models and formal tactic engines, so relevant hypotheses fall out of scope while the search space still keeps growing.
- Once that context window is saturated, the tactic search space explodes: scaling conventions, compactness topologies, and gauge choices start interacting combinatorially instead of locally.
- The architecture addresses this by compressing each edge to a bounded local reasoning problem whose hypotheses, outputs, and failure routes fit inside one audit context.
- Hallucinated estimates still occur, but they now surface as local contract violations rather than as hidden defects dispersed across a hundred-page manuscript.

4.2 LLM-proof generation and auditability

- The architecture acts as a compiler: it takes a Millennium-scale PDE problem and lowers it into a sequence of localized tactic puzzles that fit inside an LLM's active reasoning window.
- Each proof capsule is intentionally close to an undergraduate or first-year graduate exercise in local analysis, even when the surrounding theorem is globally deep.
- The architecture is therefore not only audit-friendly but search-friendly: it drastically shrinks the local tactic space that a model or proof assistant must explore.
- Axiomatizing the literature is part of the design. Established theorems such as CKN concentration, Seregin-type limits, or downstream Liouville inputs can be imported as typed capsules while the machine verifies the routing between them.
- This turns AI assistance from vague theorem imitation into structured theorem transport across a graph of explicitly declared interfaces.

4.3 Audit dimensions

Audit type	Local question
Scaling audit	Do all terms transform with the claimed critical dimension?
Sign audit	Are modulation signs consistent across the chart, PDE, energy inequality, and cost identity?
Hypothesis audit	Is each theorem invoked only after its assumptions have been generated?
Locality audit	Does the estimate use only the declared cylinder, window, or compact state?
Pressure/gauge audit	Are pressure normalizations and gauge choices compatible under limits?
Compactness audit	Is the topology strong enough for every nonlinear passage?
Retention audit	Does the retained obstruction survive this edge?
Failure-route audit	If the estimate fails, is the branch routed to a named state?
Dependency audit	Is there any circular use of a closure theorem inside its own entry route?

4.4 Capsule prompt template for LLM audit

```
Task: Audit the following proof capsule.
Input contract:
  [List exact hypotheses.]
Output contract:
  [List exact conclusion.]
Retained obstruction:
  [Name the nonzero invariant.]
Failure routes:
  [List named exits if a hypothesis fails.]
Audit checks:
  1. Verify scaling of every term.
  2. Verify sign conventions.
  3. Verify pressure/gauge normalization.
  4. Verify compactness topology and nonlinear passage.
  5. Verify no hidden global estimate is used.
  6. Verify the retained obstruction survives.
  7. Verify all dependencies are upstream.
Return:
  PASS / FAIL / NEEDS ROUTING, with exact failing line or hypothesis.
```

4.5 How fault containment helps human referees

- A referee can mark a capsule invalid without needing to reject the entire architecture.

- The graph then records a precise open node: missing estimate, unstable observable, nonexhaustive routing, or unclosed terminal stratum.
- This turns proof failure into useful diagnostic information rather than an opaque collapse.
- Allows to parallelize the repair process: different experts can work on different open nodes without stepping on each other's verified subgraph.

5 The Navier–Stokes Program as a Case Study

5.1 The case-study role

- The Navier–Stokes regularity problem is used as a stress test for the architecture because it combines scaling, pressure, compactness, and critical concentration.
- The Millennium-problem setting is ideal because it forces the architecture to confront the hardest interaction between nonlocal pressure, critical scaling, and compactness without hiding behind generic slogans.
- The case study should be read as a proof-design ledger whose analytic capsules are individually typed, auditable, and routable.
- Its value is not only explanatory: the routing graph is the logical AST that guarantees exhaustion and forbids ghost branches from escaping the local closure mechanisms.

5.2 First gate: critical concentration

- Candidate singularity implies positive scale-critical concentration after excluding the epsilon-regularity branch.
- The retained obstruction begins as positive CKN or velocity concentration.
- Vanishing concentration is closed immediately by local epsilon-regularity.

5.3 Second gate: Type I versus Type II

- Type I branch: local pointwise Type I envelope allows ancient-profile extraction.
- Type II branch: no local Type I envelope, so one studies retained local windows around moving concentration cores.
- The branches are separated because their compactness mechanisms are different.

5.4 Type I architecture

- Pipeline: concentration $\to$ no-escape $\to$ Seregin extraction $\to$ bounded centered ancient profile $\to$ class routing.
- Direct closure rows: small, stationary L^3, uniformly tight, and structure/decay-covered ancient profiles.
- The point of the routing is that each closure uses only the local profile data generated by the branch, not a whole-space bound carried from the initial solution.

- Residual row: not a wastebasket, but a terminal-dynamics object with its own descendant, tail, affine, diffuse, and endpoint routes.

5.5 Type II architecture

- Pipeline: retained core $\to$ repaired gauge $\to$ represented local equation $\to$ compactness/cost/-multibubble/cascade/terminal routing.
- Repaired gauges turn scale drift and center drift into explicit modulation coefficients rather than hidden compactness errors.
- Type II routing discards global critical norms in favor of compact-window observables, local Calderón–Zygmund pressure control, and scale-collapse cost estimates tied to the active core.
- Cost and carrier routes treat scale collapse without assuming a global critical norm or a whole-space Liouville theorem.

5.6 State-space diagram in words

```
terminal singular branch
  -> concentration gate
      -> no concentration: regularity
      -> positive concentration
          -> Type I envelope
              -> Seregin ancient state
              -> direct Liouville classes or residual class
          -> Type II branch
              -> repaired gauge / local state-space alternatives
              -> compact core, rough core, multibubble, cascade,
                 scale collapse, carrier, terminal profile, exterior exits
```

- The flow diagram is not a mere summary figure; it is the explicit AST of the proof, recording exhaustion of cases and forbidding undeclared branch escapes.
- Once a branch enters this tree, every future estimate is attached to a named local state and every terminal trap is tied to a retained obstruction.

6 Verification Ledgers

6.1 Interface ledger

Interface	Source output	Target input
Concentration gate	positive critical mass	Type I/Type II routing
Type I extraction	bounded ancient profile with retained mass	direct classes or residual class
Residual routing	realized descendant or terminal state	matching residual closure capsule
Type II entry	retained moving core	repaired-gauge representation
Gauge representation	local equation with a, b and pressure atlas	compactness, Caccioppoli, cost, and carrier capsules
Terminal profile route	retained terminal mass	profile extraction or hidden-mass exit

6.2 Retained-obstruction ledger

Obstruction	Produced by	Contradicted by
positive CKN mass	concentration gate	epsilon-regularity or vanishing pressure/velocity
compact local L^3 mass	Seregin extraction	Liouville zero conclusion
residual ancestry	descendant realization	closure of non-realized auxiliary objects is disallowed
retained Type II core mass	selected core construction	compact zero limit or finite-cost collapse
carrier cost	local cost validation	corrected monotonicity plus carrier routing
terminal critical mass	terminal profile extraction	profile exhaustion and perturbative remainder

6.3 Closure-status ledger

- Every terminal node should have status `closed`, `imported`, `open`, or `conjectural`.
- Every imported theorem should be stated with exact hypotheses and bibliographic source.
- Every open node should be allowed to remain open without invalidating unrelated closed nodes.

6.4 Repair protocol

1. Identify the failing capsule and its outgoing edge.

2. Decide whether the estimate is false, under-hypothesized, or assigned to the wrong state.
3. Add a named failure route if the negation has stable PDE meaning.
4. Split the state if two geometrically different failures were conflated.
5. Re-run the dependency audit for all downstream closures.

7 Portability Beyond Navier–Stokes

7.1 General principle

- The portable idea is not to copy the Navier–Stokes strata.
- The portable idea is to identify invariant observables, route failure modes, and use only rigidity theorems whose hypotheses have been generated.

7.2 Potential targets

Problem class	Likely retained obstruction	Likely states
Geometric flows	curvature concentration or entropy drop	tangent flows, necks, bubbles, ancient solitons
Dispersive PDE	critical mass/energy or scattering norm	profiles, soliton channels, radiation, cascades
Gauge theories	energy concentration and gauge class	bubbles, Coulomb gauges, neck regions, defect measures
Harmonic map heat flow	energy density and topological charge	bubble trees, necks, no-neck defects
Nonlinear wave equations	energy packets and channels	profiles, self-similar cones, exterior radiation

7.3 What changes in another PDE

- The critical observable changes.
- The compactness topology changes.
- The gauge or symmetry group changes.
- The closure theorems change.
- The architecture remains: retained obstruction, routing graph, proof capsules, terminal closures, fault containment.

8 Exception Handling and Proof Resilience

8.1 Dynamic repair via sub-branching

- When a reviewer or machine auditor finds an unclosed estimate, the architecture does not collapse; it instructs the proof to spawn a new named geometric stratum capturing the observed failure mode.
- A failed compactness route can become a compactness-defect branch, a failed modulation bound can become a modulation-failure branch, and a failed pressure normalization can become a pressure-atlas branch.
- The key principle is that unresolved behavior is promoted into a first-class state rather than left as an informal gap inside the trunk of the proof.
- Sub-branching is therefore not a sign of weakness but the mechanism by which the architecture converts critique into new local mathematics.

8.2 Exception handling matrix

Exception detected	Resilience response
scaling formula wrong	quarantine the chart capsule, invalidate only downstream sign-dependent edges, and preserve the rest of the routed graph
compactness lemma too strong	spawn a compactness-failure stratum or strengthen the input contract before re-entering the main branch
pressure gauge incompatible	repair the local pressure atlas or reroute to a pressure/exterior observer state
retained mass not preserved	replace the retained observable or split the branch until preservation is restored locally
Liouville theorem invoked too early	move the closure capsule downstream until its hypotheses are actually generated
cost identity wrong sign	reopen the scale-collapse branch and keep all unrelated branches certified
residual class not exhausted	mark the residual node open, attach new descendant routes, and leave upstream routing intact

9 Conclusion

- Long PDE regularity proofs should be organized as auditable routing graphs, not only as linear manuscripts.
- The decisive move is to discard hidden global estimates in favor of local state-space stratifications whose observables, gauges, and closure mechanisms remain bounded and machine-readable.

- Local proof capsules make LLM assistance stronger because every estimate is forced to declare hypotheses, outputs, retained obstructions, and failure routes inside one audit window.
- The monolithic human-readable proof is no longer the only viable endpoint for nonlinear mathematics; distributed, fault-tolerant, machine-auditable proof networks are becoming the natural format for problems of this complexity.
- By compiling continuous geometric mechanics into a discrete proof AST, the architecture bridges the gap between critical PDE analysis and automated theorem proving in systems such as Lean 4.
- The next step is to maintain a machine-readable ledger linking every capsule to its source theorem, dependency path, status, and audit history.

A Proof-Capsule Template

A.1 Minimal template

```
Capsule name:
Implementation labels:
  L1. [theorem/proposition/lemma label]
  L2. [supporting theorem/proposition/lemma label]
Input contract:
  H1. [local cylinder/window/state]
  H2. [bounds and topology]
  H3. [pressure/gauge normalization]
Output contract:
  O1. [local estimate or routing conclusion]
Retained obstruction:
  I. [quantity and survival statement]
Failure routes:
  F1. [named exit]
  F2. [named exit]
Audit checklist:
  A1. scaling
  A2. sign convention
  A3. pressure/gauge
  A4. compactness topology
  A5. nonlinear passage
  A6. retention
  A7. dependencies
Closure status:
  verified / imported / open / conjectural
```

A.2 Example capsule: selected-window compactness extraction

- Capsule name: Selected-window compactness extraction.
- Implementation labels:
- L1. `paper1:prop:compactness`.

- L2. `paper1:lem:D-localization` for the smaller-cylinder pressure bound used inside the compactness proof.
- Source: Proposition "Compactness on selected windows" in the companion manuscript *Type II Regularity for Navier–Stokes*.
- Input contract:
- H1. A represented suitable sequence on Q_{R_*} solving the local renormalized equation.
- H2. Uniform compact-cylinder bounds $A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \leq M$.
- H3. Uniformly bounded modulation coefficients a_n, b_n on the selected time interval.
- Output contract:
- O1. For every $r < R_*$, a subsequence converges weak-* in $L_t^\infty L_x^2$, weakly in $L_t^2 H_x^1$, and strongly in $L^3(Q_r)$.
- O2. Mean-normalized pressure converges weakly in $L^{3/2}(Q_r)$, and the limit still satisfies the represented equation distributionally on smaller cylinders.
- Retained obstruction:
- I. The capsule does not itself contradict the retained obstruction; it transports the branch into a compactness topology strong enough for later retention and contradiction capsules.
- Failure routes:
- F1. If the local pressure oscillation bound is unavailable, route to a pressure-row or pressure-compactness failure state.
- F2. If bounded modulation or the time-derivative control fails, route to modulation-failure or compactness-failure states rather than invoking Aubin–Lions implicitly.
- Audit checklist:
- A1. Check the pressure is normalized by spatial means on each compact ball.
- A2. Check the L^2 to L^3 interpolation and the use of the $L^{10/3}$ bound.
- A3. Check weak-* convergence for a_n, b_n is sufficient for passage to the limit in the lower-order terms.
- A4. Check the argument never upgrades local compact-cylinder data into a hidden whole-space compactness theorem.
- Closure status:
- Open as a transport capsule; it is an upstream interface rather than a terminal contradiction.

A.3 Example capsule: local single-core Type II exclusion

- Capsule name: Local single-core Type II exclusion.
- Implementation labels:
- L1. `paper1:thm:single-core-criterion`.

- L2. `paper1:lem:zero-dissipation` for the constant-limit reduction.
- L3. `paper1:lem:zero-contradiction` for the zero-limit contradiction.
- L4. `paper1:lem:nonzero-contradiction` for the bounded-window exit.
- Source: Theorem "Local single-core Type II criterion" in the companion manuscript *Type II Regularity for Navier–Stokes*.
- Input contract:
- H1. A compact single-core vanishing-dissipation branch in the precise sense of the Type II manuscript.
- H2. Upstream access to the named inputs "passage to selected-window branches," "local representation input," and "local pressure reconstruction."
- H3. Persistence of a selected retained core with positive concentration on Q_{r_*}.
- Output contract:
- O1. The branch cannot remain a local Type II sequence.
- O2. Any nonzero bounded selected-window limit is rerouted to the bounded-window row instead of remaining inside the strict selected-window branch.
- Retained obstruction:
- I. Positive retained velocity concentration on the selected core, encoded by a fixed lower bound $C_{V_n}(r_*) \geq \eta_0$ along the selected-window subsequence.
- Failure routes:
- F1. If the limit is spatially constant and nonzero, exit the strict selected-window Type II branch through the bounded-window routing row.
- F2. Multibubble, cascade, gauge-degenerate, and compactness-failure alternatives are not hidden inside the theorem; they are separate named rows outside the single-core capsule.
- Audit checklist:
- A1. Check the vanishing localized dissipation really implies a spatially constant limit on the selected compact cylinder.
- A2. Check the strong L^3 convergence on the core cylinder is exactly what forces the zero-limit contradiction.
- A3. Check the retained obstruction is contradicted only in the zero-constant case and rerouted rather than contradicted in the nonzero-constant case.
- A4. Check no hidden stationary, ancient, or self-similar Liouville theorem is invoked.
- Closure status:
- Verified terminal exclusion capsule inside the local single-core row, with named exits for behavior that leaves that row.

B LLM Audit Prompt Library

B.1 Scaling audit prompt

```
Given the change of variables and PDE below, verify every scaling exponent.
Return a table with columns: term, original scaling, transformed scaling, expected
    coefficient.
Flag missing powers or inconsistent conventions.
```

B.2 Sign audit prompt

```
Audit the sign convention for the modulation coefficients a and b.
Check consistency among: chart definition, chain rule, represented PDE,
local energy inequality, Caccioppoli estimate, and cost identity.
Return the first inconsistent formula if any.
```

B.3 Hypothesis audit prompt

```
For each theorem invocation in this proof segment, list the invoked theorem,
its hypotheses, the upstream lines that produce each hypothesis, and any missing
    hypothesis.
Classify each invocation as VALID, MISSING HYPOTHESIS, or CIRCULAR.
```

B.4 Pressure/gauge audit prompt

```
Check whether pressure is compared only modulo functions of time or spatial means.
Verify that every pressure compactness or pressure-flux step uses the declared
    normalization.
Flag any absolute pressure estimate not invariant under pressure gauge changes.
```

B.5 Retention audit prompt

```
Identify the retained obstruction entering this capsule.
Track it through subsequences, rescalings, gauge changes, and limits.
State exactly where the terminal contradiction uses the retained obstruction.
```

C Navier–Stokes Routing Graph Skeleton

C.1 Global routing

```
finite-energy suitable weak solution
  -> suppose terminal singular point exists
      -> concentration gate
          -> no concentration: epsilon-regularity closure
          -> positive concentration
```

```
          -> Type I envelope?
              yes: Seregin ancient-profile state space
              no: Type II retained local-window state space
```

C.2 Type I routing

```
Type I branch
  -> Seregin extraction
      -> small class: perturbative Liouville
      -> stationary L3 class: stationary rigidity
      -> uniformly tight class: endpoint ancient Liouville
      -> structure/decay class: specialized Liouville theorem
      -> residual class: residual terminal-dynamics graph
```

C.3 Residual Type I routing

```
residual class
  -> axisymmetric / rotational / stationary-hull / affine-parasitic strata
  -> tail strata: log-diffuse, Young defect, coherent critical tails
  -> terminal strata: nonactive, diffuse, mixed, active descendants
  -> generic residual: parasitic-free mildness + sequence L3 + endpoint Liouville
```

C.4 Type II routing

```
Type II branch
  -> select retained moving core
      -> repaired gauge representation
      -> compact single core
      -> rough core / H1 loss
      -> multibubble / cascade / gauge degeneracy
      -> scale collapse / finite cost / carrier routing
      -> terminal critical profile / exterior states
```

D Verification Ledger Templates

D.1 Closure status table template

Node	Closure mechanism	Status	Audit notes
[Node]	[Capsule]	verified/open	[Notes]

D.2 Dependency ledger template

Capsule	Uses	Produces	Downstream consumers
[Capsule]	[Inputs]	[Outputs]	[Downstream]

D.3 Retained-obstruction ledger template

Obstruction	Produced at	Preserved through	Terminal use
[Invariant]	[Produced at]	[Edges]	[Use]

E Expansion Plan for the Full Manuscript

E.1 Main text expansion targets

- Expand Section 1 into 4–6 pages with motivating examples of fragile monolithic proofs.
- Expand Section 2 into the conceptual bridge between local proof design and the elimination of whole-space critical estimates.
- Expand Section 3 into the formal mathematical core of the paper.
- Keep Section 4 short, practical, and free of hype.
- Keep Section 5 as a case study with explicit disclaimers about analytic verification.
- Put all large theorem-register material in appendices or supplementary files.

E.2 Professional style rules

- Avoid saying "LLMs prove the theorem"; say "LLMs audit local capsules."
- Avoid saying "all exits are closed" unless every closure capsule has been independently checked.
- Use "open node" instead of "gap" when the architecture intentionally records an unclosed branch.
- Use "proof capsule" only for statements with explicit hypotheses and outputs.
- Use "case study" for Navier–Stokes until the analytic proof series is independently mature.

E.3 Suggested companion artifacts

- A machine-readable JSON/YAML ledger of capsules and dependencies.
- A theorem-label register generated from the TeX source.
- A prompt library for local audits.
- A status dashboard marking capsules as verified, imported, open, or under revision.

References

[1] L. Caffarelli, R. Kohn, and L. Nirenberg. Partial regularity of suitable weak solutions of the Navier–Stokes equations. *Communications on Pure and Applied Mathematics*, 35(6):771–831, 1982.

[2] J. Leray. Sur le mouvement d'un liquide visqueux emplissant l'espace. *Acta Mathematica*, 63:193–248, 1934.

[3] L. Escauriaza, G. Seregin, and V. Sverak. $L_{3,\infty}$-solutions of Navier–Stokes equations and backward uniqueness. *Russian Mathematical Surveys*, 58(2):211–250, 2003.

[4] J. Necas, M. Ruzicka, and V. Sverak. On Leray's self-similar solutions of the Navier–Stokes equations. *Acta Mathematica*, 176:283–294, 1996.

[5] J.-P. Aubin and J. Simon. Compactness tools for evolution equations. *[Replace with exact references in final bibliography]*.

[6] [Placeholder] References on formal proof assistants, proof engineering, or LLM-assisted theorem proving should be added here only after exact bibliographic verification.